E0.1: 200-Tick Variable-Agent Compatibility

Engineering Supplement for the Agent-Scaling Pipeline

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Abstract

E0.1 stress-tested the unchanged Alem evaluator at $N \in \{1, 2, 3, 4, 6\}$ with two seeds and a requested 200-tick horizon. Provider-free scripted agents selected `Noop` and emitted deterministic broadcasts, isolating population-dependent infrastructure from language-model behavior. All ten episodes produced complete artifacts, and every structural gate passed: indexing, trajectory axes, targeted `Give` mappings, specialization cycling, broadcast accounting, action parsing, and deterministic saved-state reconstruction. The strict aggregate gate nevertheless **FAIL** because three episodes ended naturally when all agents died before tick 200: $N = 1$/seed 13000 at tick 114, $N = 4$/seed 13001 at tick 182, and $N = 6$/seed 13001 at tick 192. The inactive policy earned zero achievements, zero normalized task reward, zero monster kills, and zero player progression in every episode; its two-seed mean return was nonpositive at every population. Thus E0.1 clears the engineering compatibility blocker for all tested populations, but it does not show that every episode can reach a fixed horizon and provides no evidence about useful behavioral performance or coordination quality.

1 Question and decision rule

The purpose of E0.1 was to determine whether the canonical evaluation and artifact pipeline remains mechanically valid as the number of physical agents changes. It was not a performance experiment. The requested matrix comprised five population sizes, seeds 13000 and 13001, and at most 200 environment ticks per episode, for ten episodes total.

Two decisions are reported separately:

Structural gate.
An episode may terminate naturally, but must produce a complete canonical artifact and pass every shape, action, role, communication, replay, and provider-accounting check.

Strict horizon gate.
In addition to the structural requirements, every episode must execute all 200 requested ticks. In the runner's published summary, `passed` is the conjunction that includes this requirement.

This distinction is essential: an `environment_terminated` episode is a valid terminal trajectory, but it fails a protocol that literally requires 200 executed ticks.

2 Methods

2.1 Environment and execution

The run used `Alem-Coop-Symbolic`, Easy coordination difficulty, `god_mode=false`, soft specialization, unshared rewards, and the `baseline` communication topology. Each population used exactly one indexed fake client configuration per physical agent, one episode worker, and no debriefs or saved images. Populations ran sequentially. The evaluator's normal per-tick worker pool still scheduled one actor call per physical agent.

The specialization cycle was Warrior, Forager, Miner, repeated by agent index; its numeric representation was $(2, 1, 3)$. Consequently, the expected cycles were (2), $(2, 1)$, $(2, 1, 3)$, $(2, 1, 3, 2)$, and $(2, 1, 3, 2, 1, 3)$.

2.2 Scripted policy and communication

At scheduled actor turn t, agent i returned the valid action `Noop` and the deterministic payload

$$\texttt{E0|AGENT=}\ i\ \texttt{|TICK=}\ t\ \texttt{|STATE=ACTIVE}.$$

The response declared model ID `e0-scripted` and zero input, output, and reasoning tokens. No provider request was made. Under baseline broadcast, the expected accounting for an episode of T executed ticks was

$$M_{\text{emit}} = NT, \qquad M_{\text{deliver}} = N(N-1)T.$$

For $N = 1$, messages were recorded as emitted but had no recipient and therefore zero deliveries and zero delivered bytes.

2.3 Structural validation

For every population and seed, the harness checked:

- complete episode JSON, trajectory, debug, and saved-state artifacts;
- observation, action, reward, text, and trajectory axes consistent with the configured N and executed horizon;
- all ordered `Give to Agent` j mappings for $i \neq j$, including canonical round trips and action-slot indices;
- the expected specialization cycle;
- exact emitted-message and fan-out counts;
- action parse rate equal to 1.0;
- zero model calls, provider requests, and transport errors; and
- equality between a stable projection of the first saved state and a fresh reset at the same seed. The projection hashes the map, item map, player positions, player specializations, and coordination map.

2.4 Timing definitions

All timing values use a monotonic clock. `episode_wall_seconds` begins immediately before the main evaluator loop and ends after the final tick, before most post-episode artifact processing. `mean_tick_wall_seconds` measures from the start of a tick through incoming-message injection, state snapshots, concurrent actor calls, action parsing, broadcast routing, and `env.step`; population means are weighted by executed ticks. `max_tick_wall_seconds` is the largest such tick in either seed. `worker_round_wall_seconds` covers only submission and settlement of the concurrent scripted actor futures. Total run elapsed time also includes environment construction and compilation, reset/replay validation, serialization, and between-population overhead.

Because every episode contains one much larger maximum, the results also report a descriptive sensitivity statistic: accumulated tick time after removing exactly one maximum from each episode, divided by the remaining ticks. This is labeled "mean excluding per-episode max." It is derived from the recorded means, maxima, and tick counts; it is not a separately timed warm or steady-state benchmark. "Worker/tick" divides cumulative worker-round time by executed ticks.

3 Results

3.1 Gate outcomes

Published gate	Outcome
All populations produced complete artifacts	Yes
All structural checks passed	Yes
All requested 200-tick horizons were reached	No
Action parse rate was 1.0 in every episode	Yes
Zero model calls / provider requests / transport errors	Yes
Trajectory agent axes matched configured populations	Yes
All targeted `Give` mappings round-tripped	Yes
Specialization cycles matched	Yes
Broadcast accounting was exact	Yes
Saved-state replay projections matched	Yes
Structural decision	**PASS**
Strict decision, including full horizon	**FAIL**

Table 1: The strict result is false solely because the full-horizon gate is false. No structural or transport gate failed.

3.2 Population summary

N	Eps.	Ticks	Agent-ticks	Obs. dim.	Actions	Give maps	Deaths	Full horizon	Structural	Strict
1	2	314	314	9,476	55	0	1	No	**PASS**	**FAIL**
2	2	400	800	9,597	55	2	1	Yes	**PASS**	**PASS**
3	2	400	1,200	9,718	56	6	0	Yes	**PASS**	**PASS**
4	2	382	1,528	9,839	57	12	7	No	**PASS**	**FAIL**
6	2	392	2,352	10,081	59	30	8	No	**PASS**	**FAIL**
All	10	1,888	6,194	—	—	50	17	No	**PASS**	**FAIL**

Table 2: Population-level compatibility results. "Give maps" counts ordered giver–recipient mappings validated once per population, hence $N(N-1)$.

Observation length increased by 121 values per added agent:

$$d_{\text{obs}}(N) = 9476 + 121(N-1).$$

The action surface was 55 actions for $N = 1$ and $N = 2$, then increased by one targeted handoff slot per additional peer, reaching 59 at $N = 6$. These are interface measurements, not estimates of behavioral benefit.

3.3 Descriptive task performance

Table 3 follows the evaluator's standard summary fields. At each tick, an agent's reward is its newly earned achievement reward plus 0.1 times its change in health. The top-level `episode_return` is the mean of the physical agents' cumulative returns; this equality held exactly in all ten raw artifacts. "Mean return" below then gives the two episodes equal weight. Team achievement percentage, team achievement count, and player level are means of the corresponding terminal `user_info` values.

N	Eps.	Mean return	Avg. ticks	Team Ach. %	Team Ach. count	Player level	Full horizon	Alive-step %	Deaths / slots
1	2	-0.4500	157.0	0.00	0.00	0.00	1/2	N/A	1/2
2	2	-0.5250	200.0	0.00	0.00	0.00	2/2	95.88	1/4
3	2	-0.0333	200.0	0.00	0.00	0.00	2/2	100.00	0/6
4	2	-0.8625	191.0	0.00	0.00	0.00	1/2	78.73	7/8
6	2	-0.6833	196.0	0.00	0.00	0.00	1/2	86.48	8/12

Table 3: Source-style performance summary for the scripted `Noop` episodes. Alive-step percentage is recorded only for $N > 1$ and divides live agent-steps by configured-agent steps actually executed. "Deaths / slots" divides terminal recorded deaths by the $2N$ configured-agent episode slots. Both are descriptive exposure, not population effects.

Every terminal `Achievements/*` field, team task-reward percentage, normal-achievement count, shared-achievement count, player level, and monster kill count was zero. For $N > 1$, all recorded coordination and item-give attempt counts were also zero; the solo wrapper omits these fields. The corresponding zero success rates therefore mean no attempts rather than failed attempts. The generic evaluator `success_rate` is merely the fraction of artifacts with `done=true`; it was 100% and is deliberately omitted because it is not task success. Consequently, the nonpositive returns reflect net health change under an inactive policy rather than task accomplishment.

The raw seed-level returns appear in Table 4. They must not be used to rank population sizes: there are only two seeds, three horizons ended early, and changing N also changes mortality exposure and world dynamics.

3.4 Seed-level termination and mortality

N	Seed	Ticks	Deaths	Return	Terminal code	Emit	Deliver	Bytes	Episode wall (s)	Mean tick (s)	Max tick (s)
1	13000	114	1	-0.9000	`environment_terminated`	114	0	0	20.7634	0.1817	11.9478
1	13001	200	0	0.0000	`environment_truncated`	200	0	0	26.8550	0.1340	11.7007
2	13000	200	1	-0.8500	`environment_truncated`	400	400	12,580	34.4896	0.1719	10.1435
2	13001	200	0	-0.2000	`environment_truncated`	400	400	12,580	35.9126	0.1791	10.5244
3	13000	200	0	-0.0667	`environment_truncated`	600	1,200	37,740	47.2818	0.2356	11.4559
3	13001	200	0	0.0000	`environment_truncated`	600	1,200	37,740	47.3205	0.2361	11.4955
4	13000	200	3	-0.8250	`environment_truncated`	800	2,400	75,480	58.4060	0.2912	11.2831
4	13001	182	4	-0.9000	`environment_terminated`	728	2,184	68,568	55.1834	0.3025	12.0740
6	13000	200	2	-0.4667	`environment_truncated`	1,200	6,000	188,700	86.5491	0.4318	11.9992
6	13001	192	6	-0.9000	`environment_terminated`	1,152	5,760	181,020	85.4465	0.4439	11.7557

Table 4: Per-seed outcomes, communication, and timing. Return is mean cumulative reward per physical agent.

- `environment_truncated`: the episode reached the configured 200-tick cap.
- `environment_terminated`: the episode ended naturally after all agents died.

All 17 deaths were attributed to mob combat. There were zero deaths from starvation, dehydration, exhaustion, or friendly fire. Partial mortality did not terminate an episode: $N = 2$/seed 13000 reached the cap with one death, $N = 4$/seed 13000 with three deaths, and $N = 6$/seed 13000 with two deaths. Natural termination occurred only in the three rows where the death count equaled N.

N	Executed ticks	Emitted	Delivered	Delivery bytes	Deliveries / emission
1	314	314	0	0	0
2	400	800	800	25,160	1
3	400	1,200	2,400	75,480	2
4	382	1,528	4,584	144,048	3
6	392	2,352	11,760	369,720	5
All	1,888	6,194	19,544	614,408	—

Table 5: Broadcast accounting across both seeds. Every row exactly matches NT emissions and $N(N-1)T$ deliveries.

3.5 Communication accounting

The result makes the baseline's quadratic delivery surface explicit. At $N = 6$, 2,352 scripted emissions produced 11,760 delivered copies. Because the probe emits on every scheduled configured-agent turn, these counts remain algebraic even after individual agents die; they validate accounting, not useful information flow among living agents.

3.6 Timing

N	Episode wall total (s)	Mean episode (s)	Weighted mean tick (s)	Mean excl. per-episode max (s)	Max tick (s)	Worker/tick (ms)
1	47.6183	23.8092	0.151301	0.076475	11.9478	0.155216
2	70.4022	35.2011	0.175483	0.124436	10.5244	0.176712
3	94.6023	47.3012	0.235863	0.179381	11.4955	0.165132
4	113.5894	56.7947	0.296600	0.236695	12.0740	0.176943
6	171.9955	85.9978	0.437702	0.379037	11.9992	0.186564

Table 6: Population timing from the raw episode artifacts. Means are weighted by executed ticks; the exclusion statistic removes one maximum from each seed.

The ten episode timers summed to 498.207821 seconds. End-to-end run elapsed time was 659.272638 seconds, leaving 161.064817 seconds in construction, compilation, reset/replay validation, serialization, and orchestration outside the episode timers. The tick means increase with N, while the summed scripted worker-future time was only 0.326201 seconds across the entire matrix. The roughly 10–12 second maximum tick at every population is consistent with one-time compiled-environment overhead inside the measured loop; E0.1 was not designed as a warmed throughput benchmark. The artifacts store only the mean and maximum, not the maximum's tick index or cause, so this interpretation is not a proven attribution. The mean-excluding-maximum column ranges from 0.076475 seconds at $N = 1$ to 0.379037 seconds at $N = 6$, but no exact warm/steady-state timing is reported.

4 Interpretation and promotion decision

Engineering conclusion. The structural gate is **PASS** at all five population sizes. E0.1 found no indexing, action-space, trajectory, serialization, broadcast-accounting, replay, or provider-isolation defect. Therefore $N = 1, 2, 3, 4$ may proceed to the supported-range Source screen, and $N = 6$ clears the engineering prerequisite for a separately labeled extension.

Protocol conclusion. The strict 200-tick gate is **FAIL**. A future report must not describe this matrix as "ten 200-tick episodes": seven episodes reached the cap, while three ended naturally at 114, 182, and 192 ticks. For behavioral experiments, natural all-agent death should remain a valid outcome and be analyzed as mortality and time-to-termination rather than misclassified as an API or infrastructure failure. Comparisons must normalize cost and opportunity by actual executed agent-ticks as well as report the requested horizon.

Scientific claim boundary. The scripted `Noop` policy cannot show whether adding agents improves return, achievements, cooperation, communication efficiency, or robustness. Mortality patterns in two seeds are diagnostic only and must not be interpreted as a population-size effect.

5 Limitations

- Only two seeds were used, with a deliberately inactive policy.
- No hosted model, reasoning, prompt-context growth, or generated-message parsing was exercised.
- Broadcasts were deterministic synthetic payloads; semantic value, redundancy, and teammate response were not tested.
- The fixed 48-by-48 world, spawn geometry, mob pressure, role balance, and coordination opportunities are not controlled independently of population.
- $N = 1$ uses the solo wrapper, and $N = 6$ lies beyond the paper's documented one-to-four range; both require explicit labels in later figures.
- Timing includes cold compiled-environment effects and is unsuitable for estimating hosted-model throughput or cost.
- The deterministic replay check covers the stated stable state projection, not bitwise identity of every field or a full re-executed trajectory.

6 Provenance and audit trail

The run began at `2026-07-23T06:57:25.963656+00:00`, ended at `2026-07-23T07:08:25.236294+00:00`, and ran for 659.272638 seconds. It used Python 3.12.13 on `Linux-6.17.0-40-generic-x86_64-with-glibc2.42`. The source branch was `research/agent-scaling-recruitment-pipeline` at commit `5c4c64338b9aae9132f9cc32b6eec3d746964529`. The source-status record contained one pre-existing untracked replay, `Results/replays/nano_high_source_full_world.mp4`; it was outside the run root and was not an input to this report.

The canonical run root is `outputs/alem_eval/e0_1_agent_compatibility_200_v2`. It contains 82 files occupying 78,221,137 bytes. The source-of-truth summary `e0_results.json` has SHA-256 `6d28e885e241b4f58553291d6fa205fb982878f2575f6bdd3a24d711fda9b529`. The committed runner at the recorded source revision has SHA-256 `b2aeaca114686b547fe3b205351d3e483c4576f391194bc744dea06ed1d806a9`.

Resolved configuration SHA-256 values were:

N	`resolved_config.yaml` SHA-256
1	`2ed69835be2ef2a8e4779221e0c54f06d65cf10b48f18ca3406e87ed42762160`
2	`7895941037e47287b0fe671952dcc57fca81d66fb4d7a61edea3592338bda619`
3	`815c7d344cb690af1a5ba396838000c6c64a6ef72c032be88ed1e4ec3d393ca0`
4	`d0fbf4db8f0252c5ffc28f2fedff20345c113875bcd169f913e5b4638e17b41a`
6	`93583d0d834bf7fdc1226505b99c7e0e837dd028944ffe1202efc49300f669c6`

A corresponding reproducible invocation is:

```
uv run -extra baselines-llm -python 3.12
  python scripts/run_e0_agent_compatibility.py
  -output outputs/alem_eval/e0_1_agent_compatibility_200_v2
  -counts 1,2,3,4,6 -seeds 13000,13001 -steps 200
```

7 Conclusion

E0.1 validates variable-agent infrastructure through six agents under a provider-free 200-tick stress protocol. Its correct result is not a single undifferentiated pass or fail: the structural compatibility decision is **PASS**, while the literal full-horizon decision is **FAIL** because all-agent death naturally ended three trajectories. This evidence supports proceeding to carefully normalized behavioral screens, with deaths and actual agent-ticks treated as first-class outcomes.